

ERIC SALERNO

Independent AI Consultant — Agentic System Design · Multi-Agent Orchestration · MCP & Tool Design · LLM Integration

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Summary

Independent AI consultant designing and shipping production agentic systems — LLM integration, MCP server generation, tool-interface design, and autonomous-vs-human-in-the-loop workflow architecture. At Amperity, designed and built a two-part AI-assisted integration system (Python + Claude) — a HITL discovery pipeline and a dockerized execution engine — that cut third-party API integration time ~90% while keeping a human in the loop at the decisions that matter. The core architectural philosophy: deterministic Python for everything logic can handle cleanly; LLM only where natural-language reasoning adds value. Also developed a database-backed context persistence mechanism for sharing coherent state across distributed pipeline steps — a novel approach that emerged from a token-optimization constraint. The AI-Assisted Martech Integration Builder (see Portfolio) demonstrates the HITL and skill-delivery patterns publicly against real APIs. Brings something rare to agentic AI teams: ~30 years of engineering and product depth that shapes *which* decisions to automate, *where* to put the human gate, and *how* to know when an agent is actually working.

Core Strengths

Multi-agent orchestration & agent architecture · Autonomous-vs-HITL design · MCP servers & tool-interface design · RAG pipelines & retrieval architecture · LLM integration (Anthropic SDK) · Confidence flagging & uncertainty surfacing · Evaluation harnesses & agent testing · Enriched OpenAPI skill delivery · Prompt engineering & structured outputs · Python · FastAPI · REST API contracts · Agent spec design

Portfolio

AI-Assisted Martech Integration Builder

2026 · github.com/ericthebigsal/agent-integration · Live: ericthebigsal.github.io/agent-integration

A public demonstration of three agentic system design patterns working together — directed end-to-end: I own the architecture decisions, information-architecture calls, and domain

knowledge; an AI coding agent implemented the code, tests, and infra against that direction and my review.

- **Enriched skill delivery:** `backend/enrich.py` is a transformation pipeline that injects behavioral knowledge — API gotchas, silent failure modes, schema edge cases — directly into OpenAPI operation objects before MCP tool generation runs. Warnings travel with the tools into the LLM's context window, not into a separate doc that gets ignored.
- **Structured HITL gate:** Screen 3 surfaces skill-sourced warnings by severity, requires explicit per-risk acknowledgment, and gates tool generation until all items are reviewed. The human sees what the agent flagged, not the raw spec — a design decision about information architecture, not just UX.
- **Two-layer skill architecture:** enriched spec (Layer 1) handles what the LLM reads when using tools; companion skill files (Layer 2) handle runtime orchestration guidance for cross-object workflows, auth patterns, and known live issues. Separation of concerns for LLM tooling.
- FastAPI + GitHub Pages · live API verification against Klaviyo and HubSpot · 17-test suite

nanogpt-scifi — From-Scratch Transformer Training

2026 · github.com/ericthebigsal/nanogpt-scifi

Ongoing personal learning project applying the same directed-agent model to ML fundamentals: wrote the implementation plan and pedagogical scope (vendor nanoGPT, character-level tokenization, laptop-feasible model size), then directed an AI agent through TDD-first execution to train a 10.72M-parameter transformer from scratch (val loss 5.46 → 1.30, single Apple M3 GPU, zero cloud compute). The agent's first debugging attempt at a training stall went in circles; I redirected it to direct system evidence (`vm_stat/ps`), root-caused memory-pressure/swapping as the cause, and chose gradient accumulation over shrinking the model — the call that determined whether the result stayed a valid experiment or became a different, weaker one. A separate sleep-interruption was recovered with zero lost progress via checkpoint-resume.

Experience

Independent AI Consultant / Technical Advisor

Jun 2026 – Present · Remote · (also 2024 – 2025)

- Design and build production agentic systems: multi-agent orchestration, MCP server generation, tool-interface design, RAG pipeline architecture, and HITL workflow design — from agent spec through working, tested, deployed code.
- Advise teams on where automation earns its keep and where human judgment is structurally required — the architectural decision that determines whether an agentic system is trustworthy in production or just impressive in a demo.
- Translate ambiguous goals into agent specs with clear task decomposition, tool interfaces, retrieval strategies, and evaluation criteria.

Amperity — Technical Product Manager

- **Designed and built a two-part AI-assisted integration system** (Python + Claude) — **cutting integration development time ~90% (~4 weeks → ~2 days)** — with a deliberate Python-first architecture: deterministic Python for all operations where logic is unambiguous; LLM only where natural-language reasoning adds value. This is the decision most AI demos skip.
- **HITL discovery pipeline:** takes an OpenAPI spec and sample data (CSV/JSON/NDJSON), uses web search to understand the destination API, validates auth (OAuth 2.0, basic) with a live credential ping, and presents a column-to-endpoint mapping with confidence levels and required transformations. At any point the user can manually adjust mappings or guide via chat. Output: a config file fully specifying the integration. The confidence scoring and human override surface uncertainty rather than hiding it — the system declares what it doesn't know.
- **Dockerized execution engine:** exposes an endpoint receiving data from the source, loads the config, applies transformations, and orchestrates delivery — handling rate limiting, max payload size, callbacks, and pagination. Includes a custom-built credential vault and SQL database for config versioning and run history.
- **Database-backed context persistence:** during development explored distributing pipeline steps across separate Claude sessions as a token-optimization strategy, and built a mechanism using the SQL database to share coherent state between steps — a novel approach to context management that emerged from an engineering constraint.
- Config format validated against Amperity's production integration library: **85% compatible out of the box**; remaining 15% blocked only by unsupported auth mechanisms or SOAP-based APIs — a concrete measure of the design's generalizability.
- Built **6 mock third-party APIs (FastAPI)** as test harnesses — enabling development and validation of integration logic against credential-gated partners without live API access.
- Drove the connector SDK architecture: encoding the system's patterns into a reusable framework so the integration-building approach could scale without requiring the full tool each time.

Press Ganey — Senior Director, Technical Program Management

Mar 2022 – Aug 2024 · Greater Seattle — reported to CTO

- Led a flagship AI-powered data exchange with Epic — including **NLP pipelines applied to unstructured clinical data** — navigating the intersection of ML capability, healthcare data constraints, and partner API contracts.
- Defined the company's TPM discipline and career ladder from scratch; led a team of Principal TPMs on strategic cross-organization programs.

Accolade — Director of TPM / Technical Product Manager

Mar 2017 – May 2022 · Seattle, WA

- Built a scalable healthcare data platform and ETL/ELT pipelines (ingestion, normalization, processing, storage) underpinning core product capabilities.

Amazon — Senior Technical Program Manager

Oct 2014 – Jul 2015 · Seattle, WA

- Built ML-backed fraud and abuse prediction, detection, and mitigation systems for Amazon Coins — data pipelines and model-driven signal processing applied to a high-stakes trust problem.

Microsoft — Senior Program Manager

Jan 2004 – Oct 2014 · Redmond, WA

- Designed the initial APIs and shipped the **PlayReady DRM platform** (Windows 8, Silverlight 2/3/4) — zero-to-one platform API design with Tier-1 third-party adoption; served as PlayReady's technical representative to industry-standards bodies.
- Led Xbox One anti-piracy and hardware-security programs; built and shipped a security-focused SaaS.
- Early in tenure: product owner and PM for an internal **Microsoft Consulting Services knowledge platform** (customized SharePoint); managed a **20+ person offshore dev and test team** (Bangalore).

Appature — Technical Program Manager

Nov 2012 – Sep 2013 · Seattle, WA — SaaS startup, acquired by IMS Health

- Owned platform APIs for a Marketing Relationship Management SaaS; drove API integration into SnapLogic's visual designer.

Earlier Experience

- **Concur Technologies — Program Manager:** grew from developer to dev lead to PM; Fortune-500 customer engagements.
- **EDS / General Motors — Product Manager / Developer / Test:** early web-app developer; MCP, MCSD, MCT certified.

Education

University of Michigan — B.S., Industrial & Operations Engineering

Michigan Technological University — Engineering

Certifications

AI/ML Foundations (Neural Networks · Machine Learning · Thinking Machines) ·
Recommendation Systems with Python · Predictive Analytics & Data Mining · (legacy) MCP /
MCSD / MCT